

CSE 445 SECTION 5 GROUP 7

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Unsupervised Real-Time Multi-Object Tracking Using Motion-Based Detection and k-Means Clustering

Abstract

Detecting and tracking objects in videos is a core challenge in computer vision relevant to surveillance, smart transportation systems and the study of human activities. Although deep learning breakthroughs have produced results these methods often depend on extensive labeled datasets and significant computational power restricting their use in scenarios, with limited data and computational capacity. This study presents a framework, for detecting and tracking multiple objects by combining motion-driven foreground segmentation K-Means clustering K-Nearest Neighbors (KNN) classification and Kalman filter-driven state prediction. In a training stage object features are clustered automatically without manual labeling and the generated pseudo-labels serve to train a compact classifier. In the inference phase objects identified are. Tracked in real time employing Kalman filtering. Experimental results on real-world video sequences demonstrate that the proposed method achieves stable multi-object tracking while remaining fully explainable and computationally efficient. The framework provides a practical alternative to deep learning-based trackers in low-data settings.

1. Introduction

1.1 Background

Abstract Multi-object tracking (MOT) involves identifying objects within a video sequence and preserving their identity over time. Traditional MOT frameworks generally include object detection, data association and trajectory estimation. Initial methods utilized motion information and probabilistic filtering while contemporary techniques largely leverage neural networks, for detection and tracking.

Deep learning-based methods, such as YOLO-based detectors combined with Deep SORT-style trackers, have become dominant due to their high accuracy. However, these methods rely heavily on large annotated datasets and GPU-accelerated computation, making them less suitable for scenarios where labeled data is scarce or computational resources are limited.

1.2 Motivation

In practical scenarios—, like small-scale surveillance setups, embedded devices and specialized monitoring—extensive labeled datasets are not accessible. Furthermore, solutions relying entirely on learning frequently suffer from a lack of interpretability rendering them hard to understand or troubleshoot.

These difficulties encourage the investigation of unsupervised and traditional machine learning techniques that're capable of Operating without labeled training data, Adapt automatically to new environments and remain computationally efficient and explainable.

1.3 Challenges of Fully Deep Learning-Based Approaches in Low-Data Settings

Although they perform well deep learning-based MOT systems encounter challenges in scenarios, with limited data. Among those threats include Data reliance: Extensive labeled datasets are necessary to train models, Overfitting threat: Limited datasets result in generalization., Computational expense: Achieving real-time performance typically demands high-end GPUs., Interpretability: It is challenging to clarify model decisions. These constraints emphasize the necessity for methods that achieve a balance, between effectiveness, efficiency and clarity.

1.4 Contributions

The primary contributions of this study include the following: An unsupervised system, for detecting and tracking multiple objects that does not need any annotated data. A hybrid learning strategy combining K-Means clustering and KNN classification for adaptive object categorization. A tracking component using a Kalman filter that guarantees fluid object paths. Lastly, A modular and explainable pipeline suitable for low-data and real-time applications.

2. Related Work

Recent research has thoroughly investigated object detection and tracking using deep learning techniques. Valverde et al. [2] Created a transfer learning-driven neural network for automated glaucoma classification from color fundus images. The researchers tested CNN architectures, including VGG-19 reaching an AUC of 94%, alongside strong sensitivity and specificity. Although successful these methods depend on labeled data and are designed for particular applications.

Within the realm of gesture and sign language identification numerous researches have utilized deep learning frameworks including Detectron2, EfficientDet and various YOLO versions. These studies show results but frequently face challenges like restricted datasets, absence of cross-model evaluations and minimal implementation, on embedded systems.

The existing literature allows for different observations to be drawn: The majority of research depends on specialized datasets. Numerous studies assess one learning model without conducting a comparative evaluation. Deployment on platforms, with resources is seldom discussed.

For experiments reported further, these constraints inspire the study, which steers clear of relying on deep learning and emphasizes unsupervised learning along with probabilistic tracking.

3. Methodology

3.1 System Overview

The suggested system is composed of two stages. Training Phase (Unsupervised Learning) and Inference Phase (Detection, Classification, and Tracking). Training phase also goes by the name

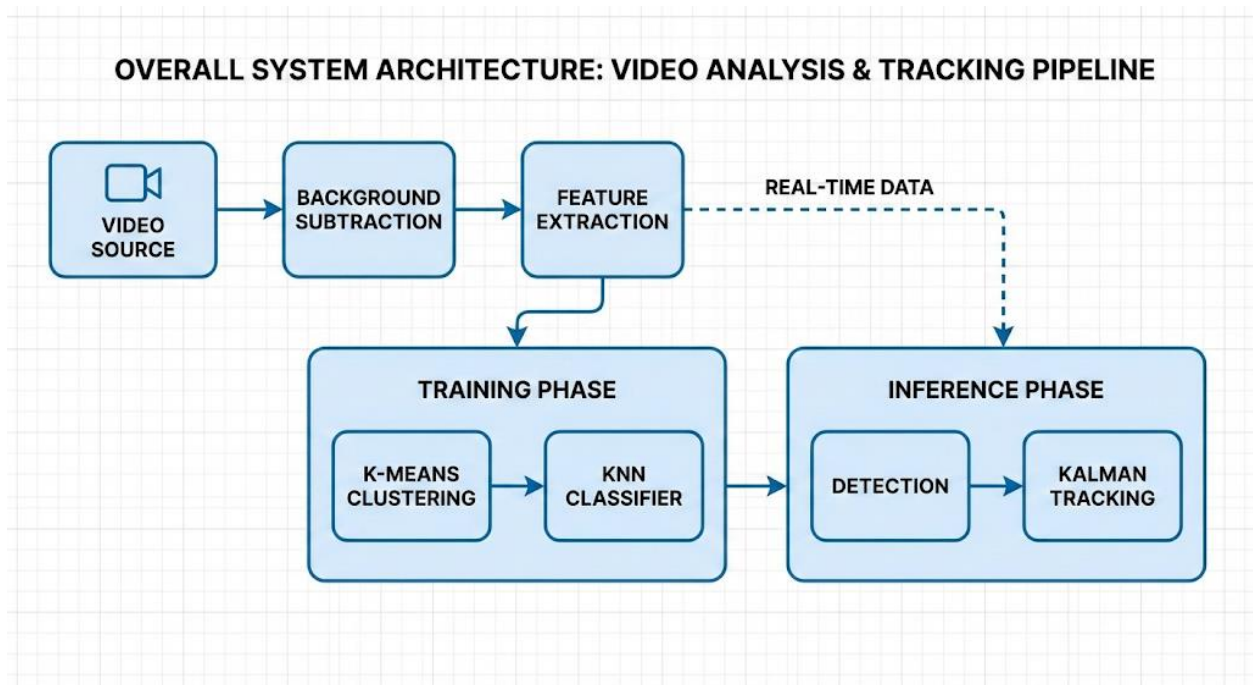


Figure 1. Block diagram of the proposed unsupervised multi-object detection and tracking framework. The system consists of two main stages: an unsupervised training phase, where object features extracted from motion cues are clustered using K-Means and used to train a KNN classifier, and an inference phase, where detected objects are classified and tracked in real time using Kalman filters.

of Learning phase. The primary objective in this stage is to find the optimum value of the model parameters which minimizes our cost. Our model has no true learning because the model does not improve representation quality over time. Inference phase or prediction phase basically runs for each frame after training. Detection, Classification and Tracking per frame, all three occur in this phase. We reuse the same features because the classifier expects the same feature space as the training phase.

3.2 Training Phase

3.2.1 Motion-Based Object Detection

Foreground objects are detected using the MOG2 background subtraction algorithm. Each video frame is converted to grayscale and smoothed using Gaussian filtering before background subtraction. Binary thresholding and morphological operations are applied to refine the foreground mask.

3.2.2 Feature Extraction

For each detected contour, a feature vector is computed as:

$$\mathbf{x} = \left[\frac{c_x}{W}, \frac{c_y}{H}, \frac{A}{WH}, \frac{w}{h} \right]$$

where (c_x, c_y) denotes the centroid, A is the contour area, w and h are bounding box dimensions, and W, H represent frame width and height.

3.2.3 Unsupervised Clustering Using K-Means

Feature vectors gathered in a warm-up phase are grouped through K-Means. The goal of clustering is to reduce the variance within clusters:

$$\arg \min_{\{\mu_k\}} \sum_{i=1}^n \| \mathbf{x}_i - \mu_{c(i)} \|^2$$

Each cluster corresponds to a discovered object category

3.2.4 KNN Classifier Training

The groupings derived from K-Means serve as pseudo-labels to train a KNN classifier. This facilitates classification in the inference stage.

3.3 Inference Phase

3.3.1 Online Detection and Classification

At inference time motion-driven detection and feature extraction take place for every frame. The KNN classifier, which has been trained assigns a class label to each object detected.

3.3.2 Kalman Filter-Based Tracking

Every object class corresponds to a Kalman filter with its state vector specified as:

$$\mathbf{x}_t = [x_t, y_t, v_{x,t}, v_{y,t}]^T$$

The state transition framework presumes motion at a velocity while the measurement framework includes the detected centroid locations. Kalman filtering enables trajectory estimation that's resilient to noise.

3.3.3 Tracker Management

Trackers that remain inactive, for a specified period are deleted to avoid lingering objects.

4. Experimental Results

Assessing -object detection and tracking frameworks generally necessitates ground-truth labels. Yet in unsupervised and real-time tracking contexts such labels are frequently missing. In this study, detection and tracking effectiveness are measured using pseudo ground truth obtained from consistent tracks. The assessment adheres to recognized conventions in -object tracking research while being appropriate for scenarios, with limited data. The evaluation of detection effectiveness employs Precision and Recall metrics that respectively measure the correctness and thoroughness of object detection.

A detected bounding box is classified as a positive (TP) when it has adequate overlap with a pseudo ground-truth object. Overlap is quantified by the Intersection, over Union (IoU) metric, which is defined as:

$$\text{IoU} = \frac{|B_d \cap B_{gt}|}{|B_d \cup B_{gt}|}$$

Where B_d denotes the detected bounding box and B_{gt} denotes the pseudo ground-truth bounding box. A detection is counted as correct if $\text{IoU} \geq 0.5$.

based on the quantities of positives (TP) false positives (FP) and false negatives (FN) precision and recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}$$
$$\text{Recall} = \frac{TP}{TP + FN}$$

Precision indicates the ratio of identified objects that match objects while recall gauges the ratio of true objects that are correctly identified.

To assess the tracking efficacy this study uses Multiple Object Tracking Accuracy (MOTA) a commonly adopted metric in multi-object tracking evaluations, like MOTChallenge. MOTA simultaneously penalizes three categories of tracking mistakes: False Negatives (FN): missed detections, False Positives (FP): spurious detections, and Identity Switches (IDSW): incorrect reassignment of object identities over time. MOTA is described as:

$$MOTA = 1 - \frac{FN + FP + IDSW}{GT}$$

Where GT represents the total number of ground-truth object instances. In the proposed unsupervised framework, GT is estimated based on the count of consistent pseudo ground-truth tracks. A greater MOTA score signifies overall tracking accuracy with a score of 1 denoting flawless tracking. In the absence of ground truth pseudo ground truth is generated by leveraging stable Kalman filter tracks. Objects lasting beyond a set threshold of frames are considered as ground-truth examples. This approach has been used in unsupervised tracking research and offers a fair estimate, for assessing relative system effectiveness. While pseudo ground truth does not replace manually annotated data, it enables consistent and reproducible evaluation of unsupervised tracking systems. Precision and recall capture detection reliability, while MOTA provides a holistic measure of tracking accuracy by accounting for detection and identity consistency errors. As pseudo ground truth does not replace manually annotated data, consistency of evaluation becomes our most promising result from the experiment.

4.1 Experimental Setup

The suggested framework was tested on real-world videos featuring moving entities, like vehicles and pedestrians. Every experiment was performed on a CPU- system without the use of GPU acceleration.

4.2 Qualitative Results

The system effectively monitored several objects, in real time. Consistent object identities were preserved between frames and Kalman filtering reduced noise in the detections. This is because of

Example of Kalman filter for tracking a moving object in 1-D

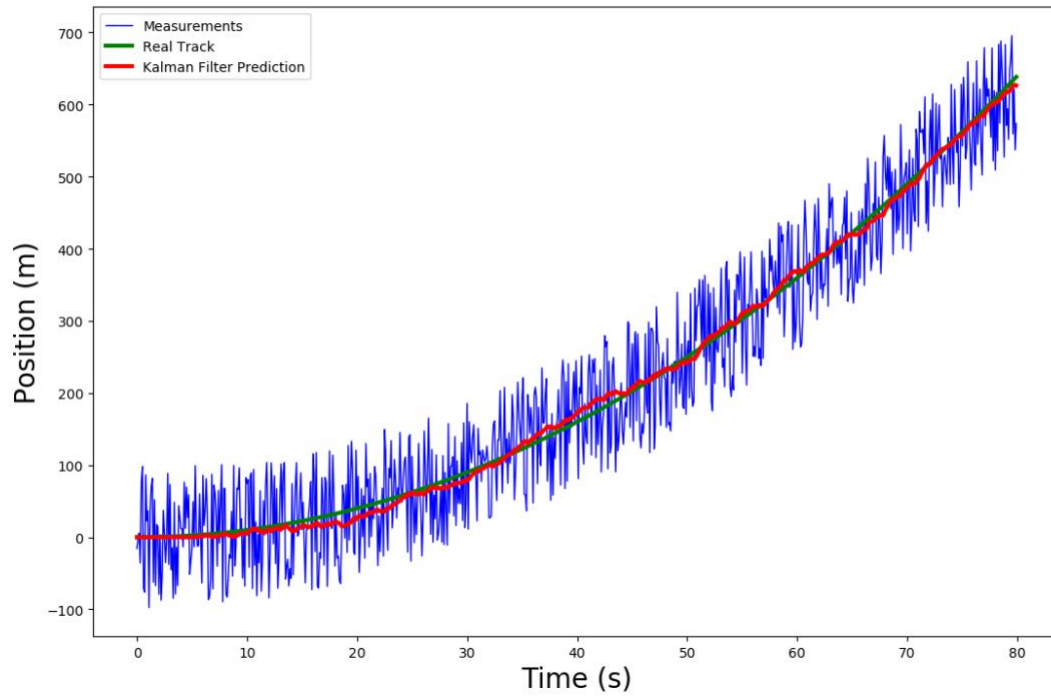


Figure 2. Time-Position graph of real tracking a moving object using Kalman Filter in 1-D. Kalman filter-based tracking ensures temporally smooth trajectories and identity consistency across frames despite noisy detections.

bayesian state-estimation that object continuity is maintained and any short occlusions are also managed.

5. Discussion

The experimental findings indicate that unsupervised learning paired with tracking methods can provide dependable multi-object tracking in settings with limited data. The suggested approach delivers clarity and flexibility rendering it appropriate, for use in situations where deep learning is not feasible.

Nevertheless, effectiveness relies on the accuracy of background elimination and the distinctness of features. Crowds that are extended occlusions continue to pose difficulties.

6. Conclusion and Future Work

This study introduced a framework for detecting and tracking multiple objects by applying K-Means clustering KNN classification and Kalman filtering. The approach functions without the need, for labeled datasets maintains efficiency and delivers easily understandable tracking outcomes.

The main limitation of the project comes from the use of hybrid learning. K-Means presumes clusters to be spherical which might not effectively represent distributions of object features. Cluster drift can happen when the visual characteristics of objects evolve over time. The cost of KNN increases, as the amount of data expands. Latency grows as more samples gather unless you trim. A Cluster is not equal to class and clusters do not necessarily correspond to a semantic object category. There is no true learning in this model doesn't improve representation quality over time.

In the future, we would implement adaptive or incremental clustering techniques along with the integration of appearance descriptors for re-identification. It would be also interesting to extend to non-linear motion models and deployment on embedded edge devices.

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